

This assignment will not be graded and is only for practice.

Level 1

1) Consider the following statements:

Statement 1: $0(0, 0) + 0(1, 1) = (0, 0)$ implies that the set $\{(0, 0), (1, 1)\}$ is a linearly independent subset of $\mathbb{R}^2$.

Statement 2: $2(1, 0) + 2(0, 1) = (2, 2)$ implies that the set $\{(1, 0), (0, 1), (2, 2)\}$ is a linearly dependent subset of $\mathbb{R}^2$.

Statement 3: $2(1, 0) + 2(0, 1) = (2, 2)$ implies that the set $\{(1, 0), (2, 2)\}$ is a linearly dependent subset of $\mathbb{R}^2$.

Find the number of correct statements.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

2) If $\alpha(1, 1, 0) + \beta(0, 1, 1) + \gamma(1, 0, 1) = v$ for some $\alpha, \beta, \gamma \in \mathbb{R}$ and $v \in \mathbb{R}^3$, then choose the set of correct options. **1 point**

- ☐ If $v = (0, 0, 0)$, then one of the possible solutions of α, β , and γ is $\alpha = \frac{1}{2} = \gamma$, and $\beta = -\frac{1}{2}$.
- ☐ If $v = (0, 0, 0)$, then one of the possible solutions of α, β , and γ is $\alpha = \beta = \gamma = 0$.
- ☐ If $v = (0, 0, 0)$, then the possible solution of α, β , and γ is unique.
- ☐ If $v = (1, 0, 0)$, then one of the possible solutions of α, β , and γ is $\alpha = \frac{1}{2} = \gamma$, and $\beta = -\frac{1}{2}$.
- ☐ The set $\{(1, 1, 0), (0, 1, 1), (1, 0, 1)\}$ is linearly independent.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If $v = (0, 0, 0)$, then one of the possible solutions of α, β , and γ is $\alpha = \beta = \gamma = 0$.

If $v = (0, 0, 0)$, then the possible solution of α, β , and γ is unique.

If $v = (1, 0, 0)$, then one of the possible solutions of α, β , and γ is $\alpha = \frac{1}{2} = \gamma$, and $\beta = -\frac{1}{2}$.
The set $\{(1, 1, 0), (0, 1, 1), (1, 0, 1)\}$ is linearly independent.

3) Consider a set of vectors $S = \{(1, 2), (-1, 2), (3, 1), (-3, 1)\}$ in $\mathbb{R}^2$. Choose the set **1 point** of correct options.

- ☐ $\{(1, 2), (-1, 2)\}$ is a linearly dependent set.
- ☐ $\{(-1, 2), (-3, 1)\}$ is a linearly independent set.
- ☐ $\{(-3, 1), (-1, 2), (3, 1)\}$ is a linearly dependent set.
- ☐ $\{(3, 1), (-3, 1)\}$ is a linearly dependent set.
- ☐ $\{(3, 1), (1, 2), (-3, 1)\}$ is a linearly dependent set.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(-1, 2), (-3, 1)\}$ is a linearly independent set.
 $\{(-3, 1), (-1, 2), (3, 1)\}$ is a linearly dependent set.
 $\{(3, 1), (1, 2), (-3, 1)\}$ is a linearly dependent set.

4) Let V be a vector space and non zero vectors v_1, v_2, v_3 , and $v_4 \in V$. If v_1 is a linear combination of v_i , for $i = 2, 3, 4$, then which of the following is (are) correct? **1 point**

- ☐ The set $\{v_1, v_2, v_3, v_4\}$ cannot be linearly independent.
- ☐ The set $\{v_2, v_3, v_4\}$ must be linearly dependent.
- ☐ The set $\{v_2, v_3, v_4\}$ must be linearly independent.
- ☐ The set $\{v_1\}$ must be linearly independent.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The set $\{v_1, v_2, v_3, v_4\}$ cannot be linearly independent.

The set $\{v_1\}$ must be linearly independent.

5) Choose the correct set of options.

1 point

- ☐ Union of two distinct linearly independent sets may be linearly independent.
- ☐ Union of two distinct linearly independent sets must be linearly dependent.
- ☐ Non-empty intersection of two linearly independent sets must be linearly independent.
- ☐ Non-empty intersection of two linearly independent sets must be linearly dependent.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Union of two distinct linearly independent sets may be linearly independent.

Non-empty intersection of two linearly independent sets must be linearly independent.

6) If vectors v_1, v_2, v_3 are linearly independent and $\alpha v_1 + \beta v_2 + \gamma v_3 = 0$, then find the value of $\alpha + \beta + \gamma$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 0

1 point

7)
Consider the set of matrices $S = \left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}, \begin{bmatrix} 7 \\ 8 \\ x \end{bmatrix} \right\} \subset M_{3 \times 1}(\mathbb{R})$. If $x \in \{9, 4, 2, 0\}$, then find the maximum value of x so that S is linearly independent.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 4

1 point

Level 2

8) Let S be a linearly independent subset of $\mathbb{R}^3$. Choose the correct set of options. **1 point**

- ☐ $S \cup \{(1, 0, 0)\}$ must be linearly independent.
- ☐ $S \cup \{v\}$ must be linearly independent for any $v \in \mathbb{R}^3$.
- ☐ $S \setminus \{v\}$ must be linearly independent for any $v \in \mathbb{R}^3$.
- ☐ There may exist some $v \in \mathbb{R}^3$, such that $S \cup \{v\}$ is still linearly independent.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$S \setminus \{v\}$ must be linearly independent for any $v \in \mathbb{R}^3$.

There may exist some $v \in \mathbb{R}^3$, such that $S \cup \{v\}$ is still linearly independent.

9) Consider a set of vectors $S = \{(1, 1, -1), (-1, 1, 1), (0, \frac{1}{2}, 0), (0, 1, -2), (1, 0, -2)\}$. Choose the set of correct options. **1 point**

- ☐ The singleton set $\{(0, 1, -2)\}$ is linearly dependent.
- ☐ If $\alpha, \beta \in S$ and α, β are distinct then $\{\alpha, \beta\}$ is a linearly independent set of vectors.
- ☐ The set $\{(-1, 1, 1), (0, \frac{1}{2}, 0), (0, 1, -2)\}$ is a linearly independent set of vectors.
- ☐ The set S is a linearly independent set of vectors.
- ☐ The set $\{\alpha, \beta, \gamma\}$ is a linearly dependent set of vectors for any $\alpha, \beta, \gamma \in S$, where all the three are distinct vectors.
- ☐ The set $\{\alpha, \beta, \gamma, \delta\}$ is a linearly independent set of vectors for any $\alpha, \beta, \gamma, \delta \in S$, where all the four are distinct vectors.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If $\alpha, \beta \in S$ and α, β are distinct then $\{\alpha, \beta\}$ is a linearly independent set of vectors.
 The set $\{(-1, 1, 1), (0, \frac{1}{2}, 0), (0, 1, -2)\}$ is a linearly independent set of vectors.

10) Let S be the solution set of a system of homogeneous linear equations with 3 variables and 3 equations, whose matrix representation is as follows:

1 point

$$Ax = 0$$

where A is the 3×3 coefficient matrix and x denotes the column vector $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$. Choose the set of correct options.

- ☐ If v_1 and v_2 are in S , then any linear combination of v_1 and v_2 will also be in S .
- ☐ The set S will be a subspace of $\mathbb{R}^3$, with respect to usual addition and scalar multiplication as in $\mathbb{R}^3$.
- ☐ The set $\{v_1, v_2, v_1 - v_2\}$ is a linearly dependent subset in S .
- ☐ The set $\{v_1, v_2\}$ is a linearly independent subset in S if v_1 is not a scalar multiple of v_2 .

No, the answer is incorrect.

Score: 0

Accepted Answers:

If v_1 and v_2 are in S , then any linear combination of v_1 and v_2 will also be in S .

The set S will be a subspace of $\mathbb{R}^3$, with respect to usual addition and scalar multiplication as in $\mathbb{R}^3$.

The set $\{v_1, v_2, v_1 - v_2\}$ is a linearly dependent subset in S .

The set $\{v_1, v_2\}$ is a linearly independent subset in S if v_1 is not a scalar multiple of v_2 .

Your score is: 0/10